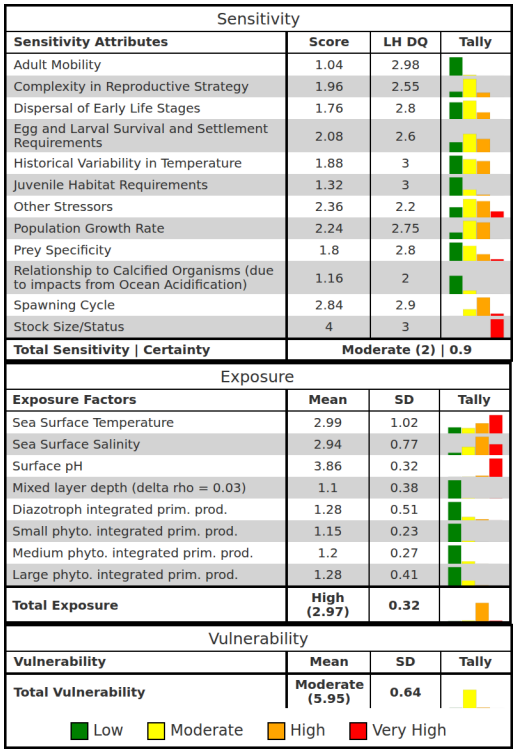
Atlantic herring (*Clupea harengus*)

Summary Scores

- **Sensitivity:** Moderate
- **Exposure:** High
- **Vulnerability:** Moderate



Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.04	2.98	
Complexity in Reproductive Strategy	1.96	2.55	
Dispersal of Early Life Stages	1.76	2.8	
Egg and Larval Survival and Settlement Requirements	2.08	2.6	
Historical Variability in Temperature	1.88	3	
Juvenile Habitat Requirements	1.32	3	
Other Stressors	2.36	2.2	
Population Growth Rate	2.24	2.75	
Prey Specificity	1.8	2.8	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.16	2	
Spawning Cycle	2.84	2.9	
Stock Size/Status	4	3	
Total Sensitivity | Certainty	**Moderate (2) | 0.9**		

Exposure

Exposure Factors	Mean	SD	Tally
Sea Surface Temperature	2.99	1.02	
Sea Surface Salinity	2.94	0.77	
Surface pH	3.86	0.32	
Mixed layer depth (delta rho = 0.03)	1.1	0.38	
Diazotroph integrated prim. prod.	1.28	0.51	
Small phyto. integrated prim. prod.	1.15	0.23	
Medium phyto. integrated prim. prod.	1.2	0.27	
Large phyto. integrated prim. prod.	1.28	0.41	
Total Exposure	**High (2.97)**	**0.32**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Moderate (5.95)**	**0.64**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Atlantic herring. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Low	Moderate	Increase
Exposure	High	High	None
Vulnerability	Low	Moderate	Increase
Directional Effect	Negative	Negative	None

Stocks

Atlantic herring is managed as a single stock or stock information was not considered for this iteration of the CVA. ## Score Descriptions:

Sensitivity:

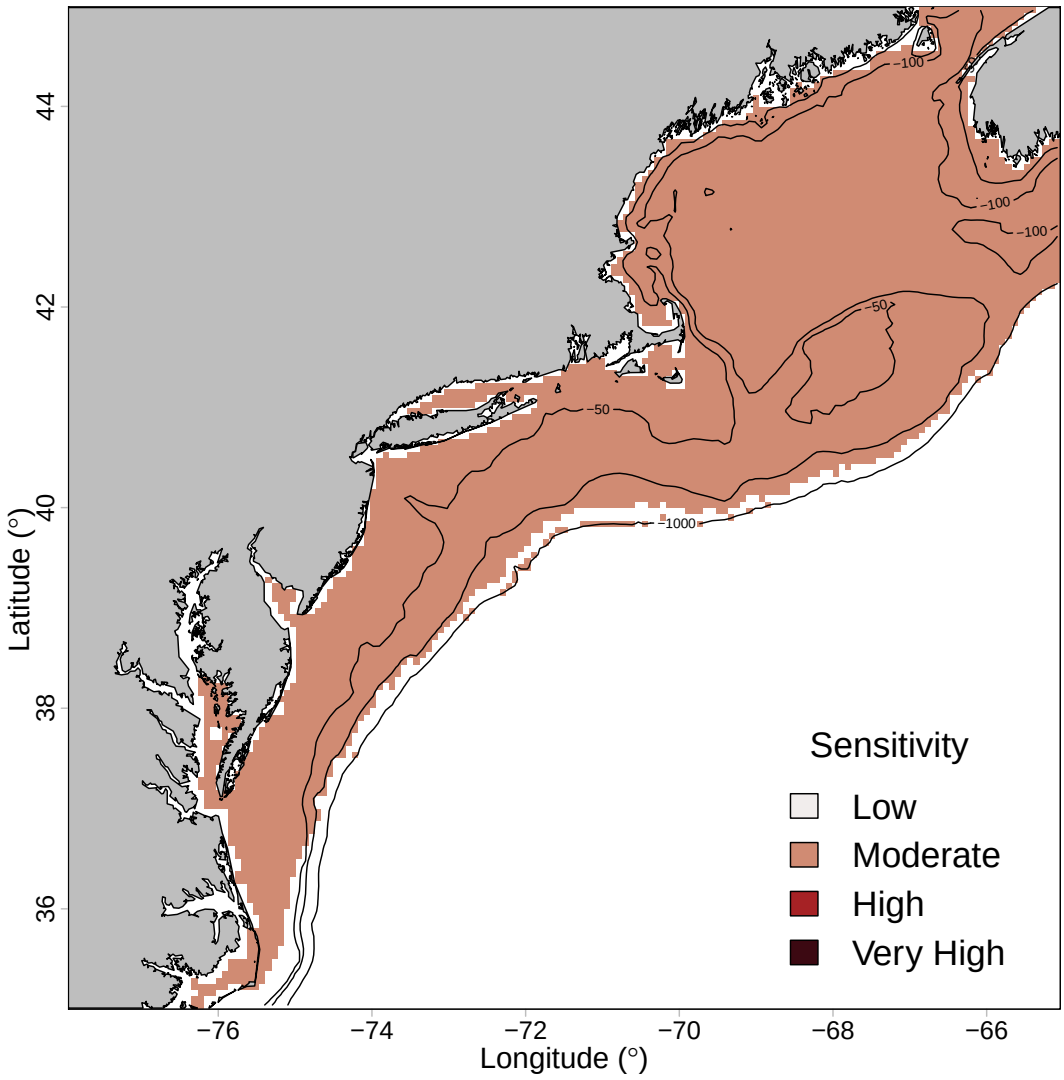


Figure 2: Atlantic herring Sensitivity on the Northeast Shelf

Atlantic herring had a **Moderate** sensitivity score, with a certainty of 90%. This score was driven by two attributes: Stock Size (4) and Spawning Cycle (2.8) (Figure 1).

Spawning success and recruitment is highly correlated with temperatures. The optimal spawning temperature window has shortened over the past 30 years (Kanwit and Libby 2009; Molina 2025). Mismatches between larvae and prey abundance lead to higher mortality (Randall et al. 2022). As of the 2024 assessment, herring are overfished but not experiencing overfishing (Northeast Fisheries Science Center (NEFSC) 2024).

In the previous assessment, herring received a Low sensitivity score. This score was driven by Early Life History Requirements, citing the need for demersal spawning habitat (Hare et al. 2016).

Exposure:

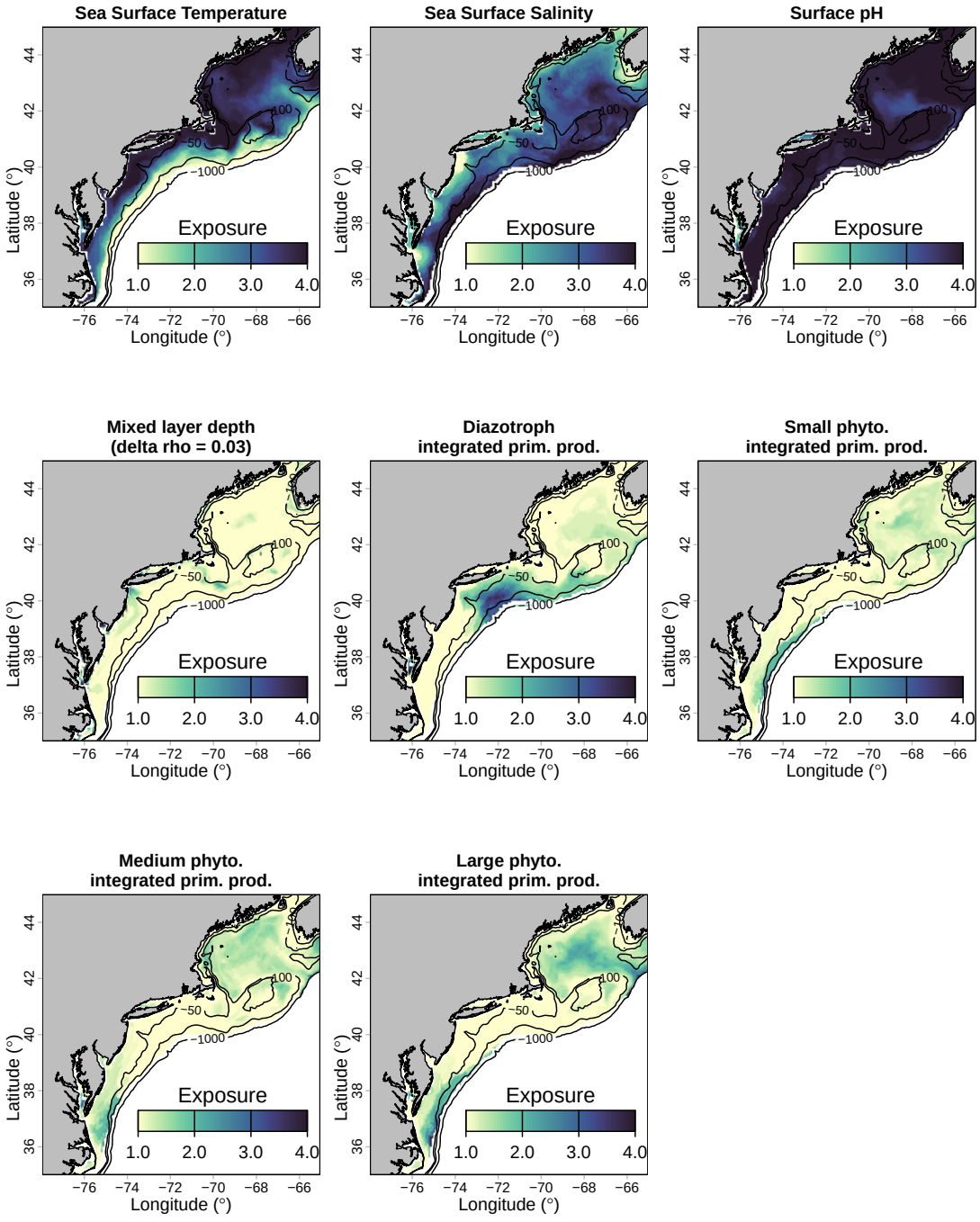


Figure 3: Maps of each Exposure Factor for Atlantic herring. Exposure was calculated between 2014-2023 and 2025-2035

Mean total exposure for Atlantic herring was **High** (2.97 +/- 0.32) (Figure 1). Exposure was uniform across the continental shelf, with Very High exposure on Georges Bank, and Low to Moderate Exposure

south of Cape Cod and east of the Chesapeake Bay (Figure 4). These patterns were driven by very high exposure due to sea surface temperature, salinity, and pH (Figure 3).

In the previous Assessment, Atlantic Herring also received a High exposure score, driven by surface temperatures and ocean acidification (Hare et al. 2016).

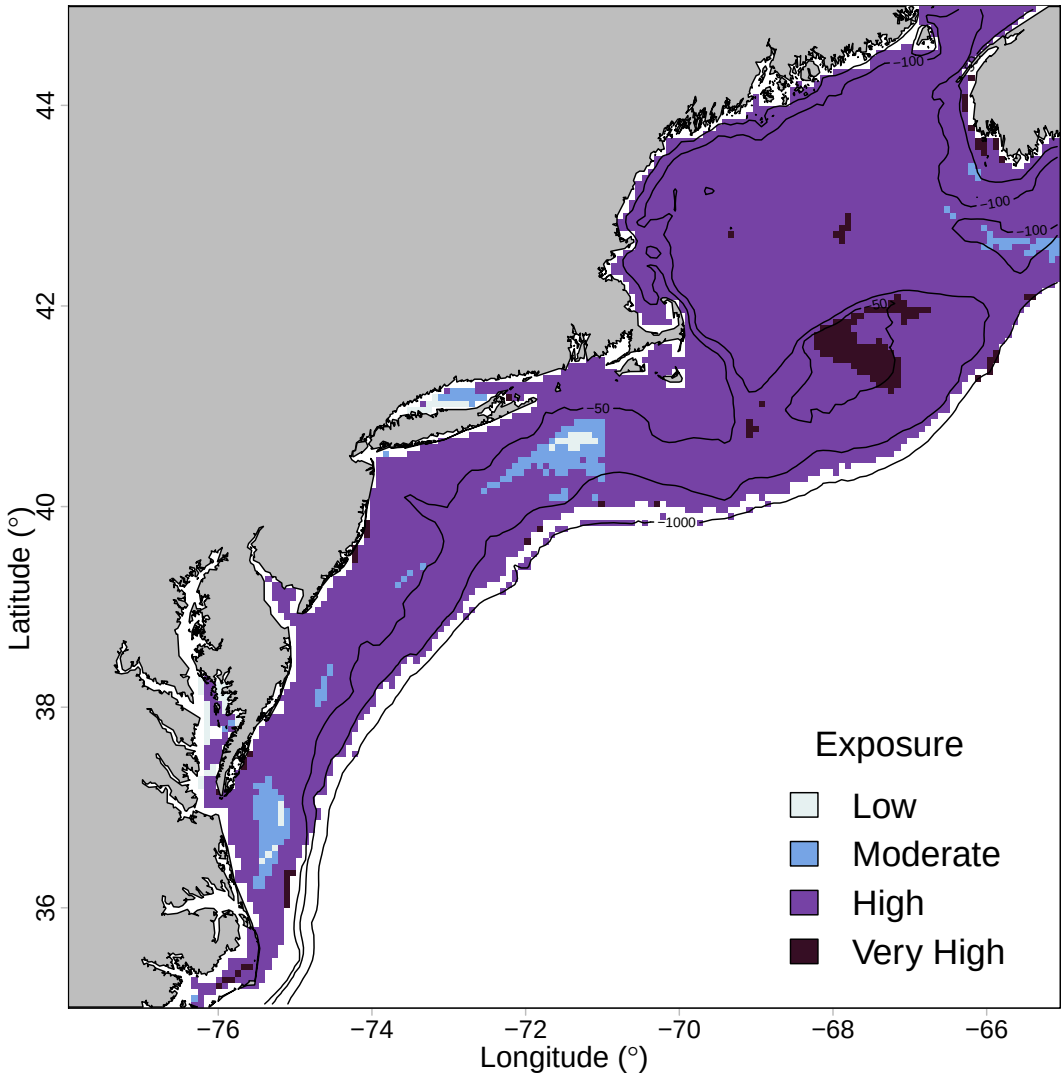


Figure 4: Total Exposure on the Northeast Shelf for Atlantic herring. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

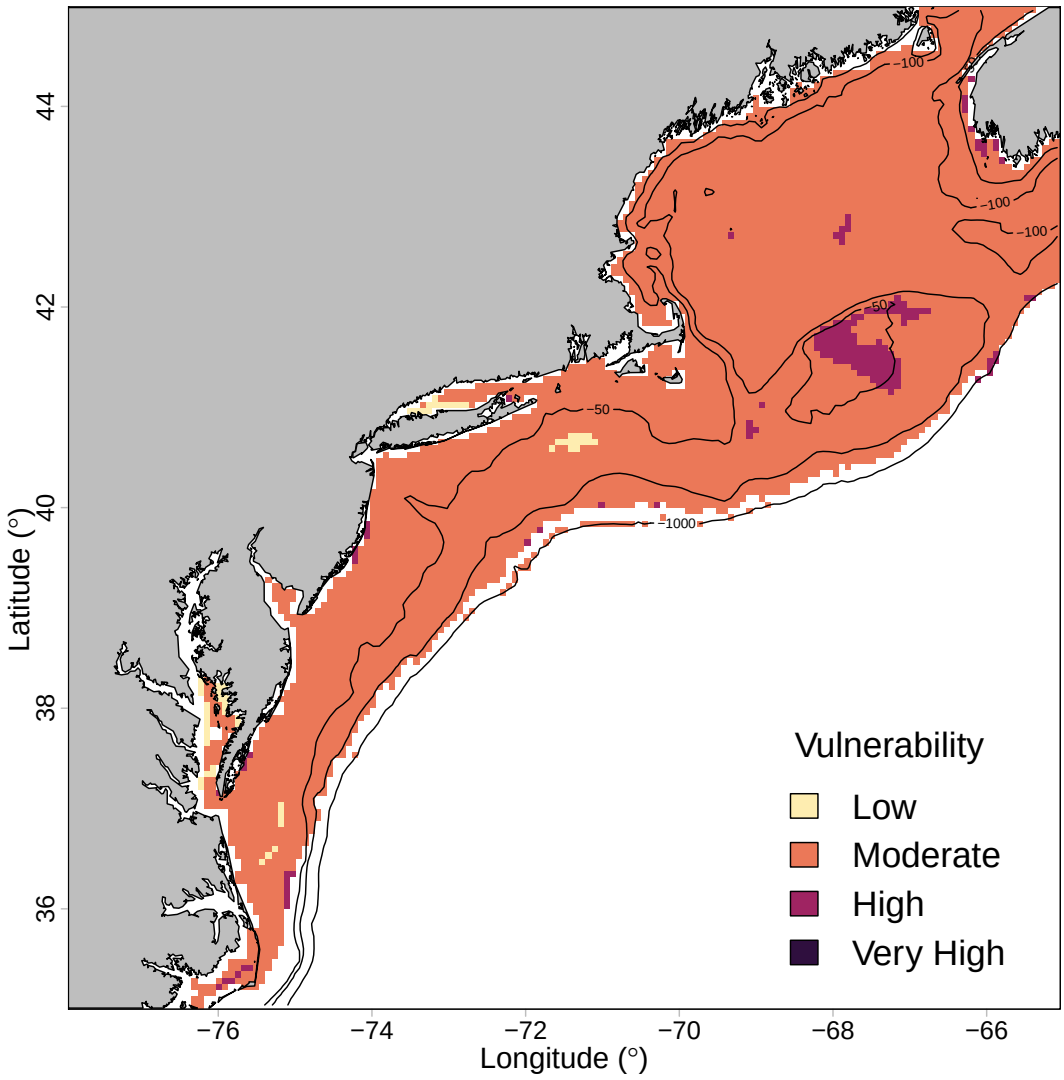


Figure 5: Vulnerability on the Northeast Shelf for Atlantic herring. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean Vulnerability for Atlantic Herring was **Moderate** (5.95 +/- 0.64) (Figure 1). Vulnerability was uniform across the shelf when all factors were considered, with peaks of High Vulnerability off Georges Bank (Figure 5).

In the previous Assessment, Herring received a Low Vulnerability Score. The change in this assessment is due to an increase in the Sensitivity Score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

95% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified for this species (Figure 1).

Model Confidence Score:

The herring ensemble model received a model confidence score of 1.6 (+/- 0.3), indicating low confidence in the model. While the model results represented general distribution patterns, experts noted that residuals (differences between observed and predicted values) were high in March and September. This was likely due to herring catchability in the surveys and interannual variability in migration timing. The ensemble model's AUC was 0.73, indicating fair performance.

Additional Metrics

Distribution Shifts:

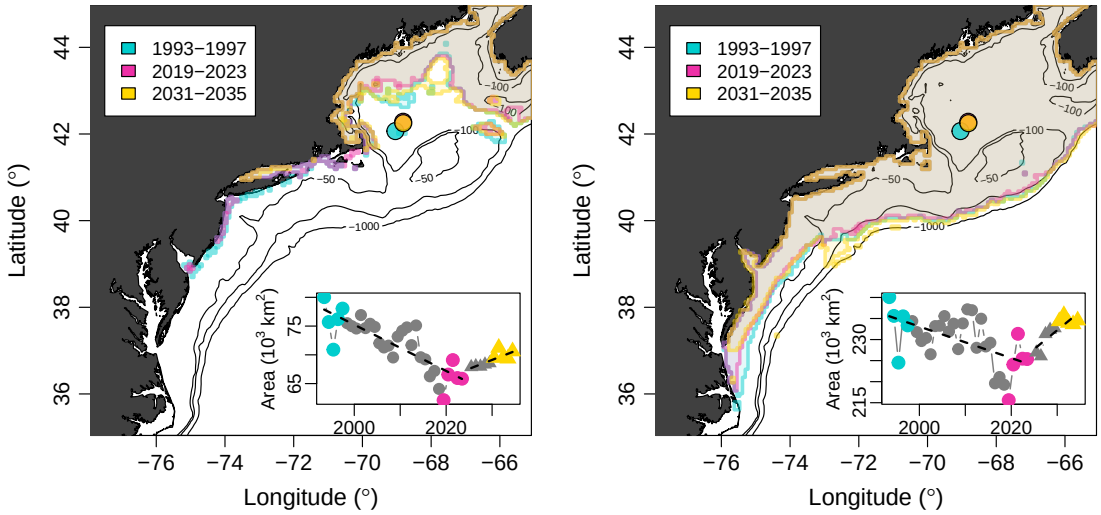


Figure 6: Distribution shifts on the Northeast Shelf for Atlantic herring. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

The weighted distribution centroid for Atlantic herring has shifted northward by 30 km since 1993. This is due to the northward shift of the southern edges of herring core habitat and their entire range. As a result, herring habitat area has declined significantly between 1993 and 2023 (Figure 6).

However, herring habitats are predicted to expand slightly in the next decade. Habitat areas are expected to significantly increase between 2025 and 2035. Core habitat in the Gulf of Maine is predicted to expand southward, resulting in a nearly 4 km southward shift in the predicted distribution weighted centroid (Figure 6).

Herring received a High Distributional Vulnerability Rank in the previous assessment due to high mobility and larval duration (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	76,345	234,718	6.11	9.1
2019-2023	30.3 km NE	65,973 (-10,372)	224,811 (-9,907)	5.67 (-0.44)	8.82 (-0.32)
2031-2035	3.7 km S	70,052 (+4,078)	234,504 (+9,693)	5.74 (+0.06)	8.65 (-0.16)

Directional Effect:

Herring are very likely to be **negatively** impacted by future climate change along the Northeast Shelf (100% certainty in expert scores). Warming temperatures and increasing bottom salinities are decreasing the amount of favorable habitat for herring (Suca et al. 2021). Mismatches between larvae and prey abundances have resulted in slower growth and higher larval mortality (Randall et al. 2022). The 2016 Climate Vulnerability Assessment for the Northeast Shelf also suggested that herring would be negatively impacted, but with more uncertainty (Hare et al. 2016).

Climate Effects on Abundance and Distribution

A number of Atlantic herring life history parameters are linked to drivers of climate change. Climate-related mismatches between Atlantic herring larval feeding and prey abundance have resulted in slower growth and higher mortality (Randall et al. 2022). Gröger et al. (2010) found that lags in the North Atlantic Oscillation (NAO) and Atlantic Multi-decadal Oscillation (AMO) were the primary drivers of Atlantic herring recruitment. Winter NAO has also indirectly influenced recruitment in multiple herring stocks (Axenrot and Hansson 2003; Oskarsson and Taggart 2010; Lynam et al. 2005) and may be used to predict recruitment with lead times up to 5 years (Gröger et al. 2010). Atlantic herring recruitment is also influenced by larval temperatures, the Gulf Stream Index, and food availability from water mass movement (Molina 2025). Favorable habitat for Atlantic herring in the Northwest Atlantic is currently decreasing with rising temperatures and bottom salinity (Suca et al. 2021), with herring populations in warmer waters responding more negatively to increasing temperatures than those in colder waters (Beaudry-Sylvestre et al. 2024). Herring larvae in the Baltic Sea are facing more heat stress and cardiac dysfunction with an increase in the number of marine heat waves (Gröger et al. 2024). Migration and distribution of Atlantic herring is also linked with thermal oceanic fronts between colder, less saline continental shelf water and warmer, more salient slope water (where plankton are also usually more abundant) (Kelly and Moring 1986).

Life History Synopsis

Atlantic herring in the Northwest Atlantic reside from southern Labrador in Canada to Cape Hatteras, NC (Stevenson and Scott 2005). They spawn primarily on Jeffreys Ledge and Georges Bank between July and November, with peaks in August - September in the Gulf of Maine. However, recent research has suggested that the optimal spawning temperature window has shortened over the past 30 years (Kanwit and Libby 2009; Molina 2025). Early life history development is inversely related to water temperature, with slower growth occurring at higher temperatures (Stevenson and Scott 2005). Eggs hatch after 6-15 days, and larval development time ranges from 4-8 months. Larvae absorb their yolk sac within 2.5 - 6 days, at which point the larvae become pelagic and can be transported away from or retained near spawning habitats, depending on prevailing current systems (Lough and Cohen 1982). Predation of eggs is linked to high mortality. Predators include haddock, cod, mackerel, winter flounder, moon snails, hermit crabs and starfish. Strong haddock recruitment has been linked to low abundances of larval herring (Boyar et al. 1973; Burbank et al. 2023). Larval herring prey on copepods, with size depending on the size of the larvae (Sherman and Honey 1971). Juveniles form large schools in coastal waters (i.e. estuaries and embayments) throughout the Gulf of Maine in the spring and summer, and overwinter in deep bays or offshore (Stevenson and Scott 2005). Juveniles and adults are planktivorous and opportunistic feeders. As they grow and the size of their jaws increases, they consume larger organisms. Their diet therefore varies with season, their age and size, and location. (Stevenson and Scott 2005; Sherman and Perkins 1971). A lack of availability of preferred prey in adults (due to strong mixing in the water column) can lead to suboptimal feeding and slow growth resulting in increased mortality from starvation and predation (Stevenson and Scott 2005). Adult herring make extensive seasonal migrations between summer spawning grounds on Georges Bank and in the Gulf of Maine and overwintering areas in southern New England and the Mid-Atlantic region (Stevenson and Scott 2005). There are two major spawning stocks in the Northwest Atlantic: the Gulf of Maine and Georges Bank/Nantucket Shoals. Stocks are separated based on larval distribution patterns, spawning times/locations, and growth rates (Iles and Sinclair 1982; Boyar et al. 1973; Anthony and Waring 1980). Both stocks are currently assessed in the U.S. as a single coastal stock complex managed by the New England Fishery Management Council. As of the most recent assessment (Northeast Fisheries Science Center (NEFSC) 2024), the stock is overfished but not experiencing overfishing based on the biological reference points for spawning stock biomass (SSB) and fishing mortality (F).

Supplemental Scores

Exposure Timeseries

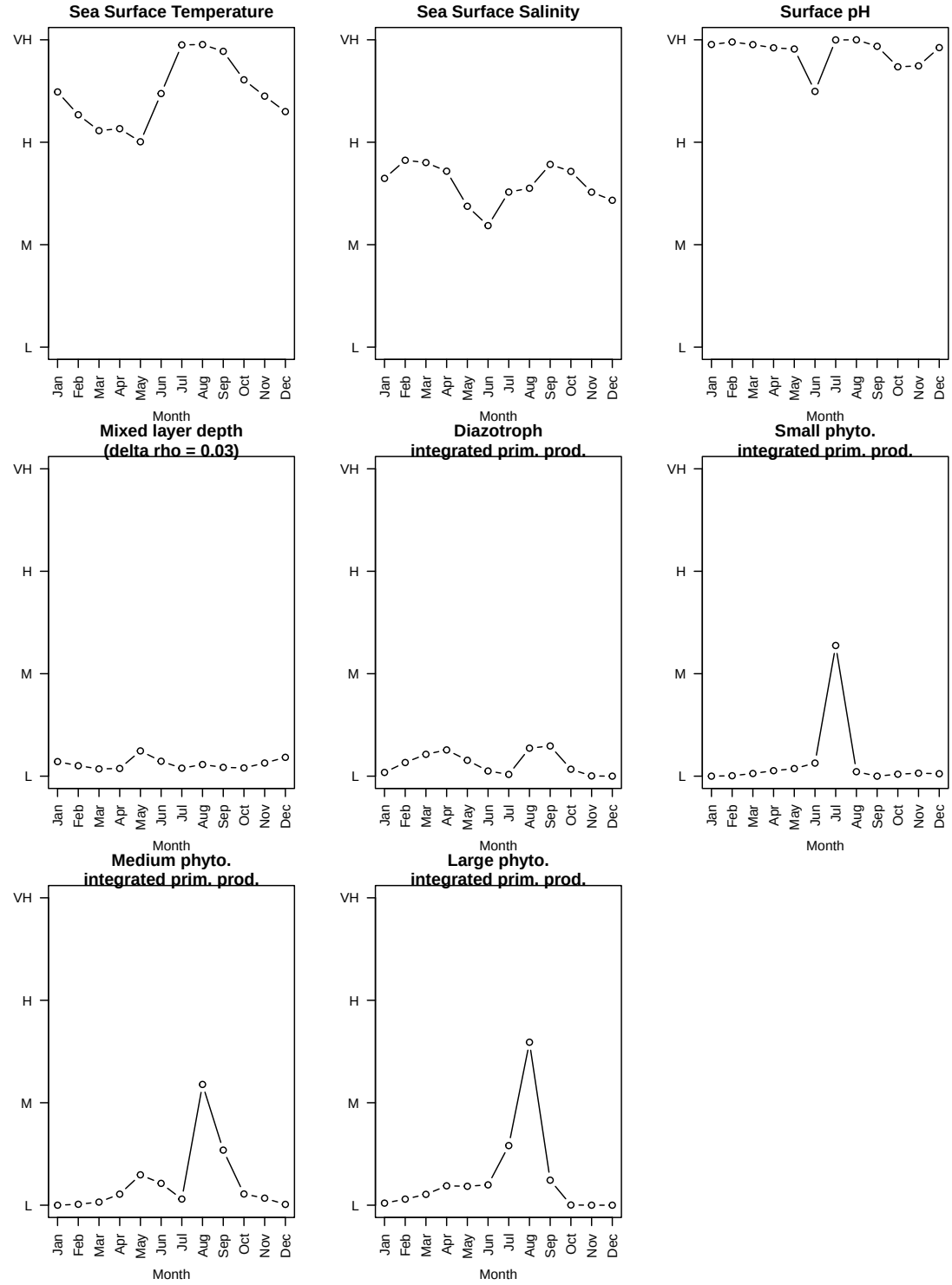


Figure 7: Monthly Timeseries of each Exposure Factor for Atlantic herring. Exposure was calculated between 2014-2023 and 2025-2035

Monthly exposure was consistently High. There was no variation across months (Figure 8; Figure 7).

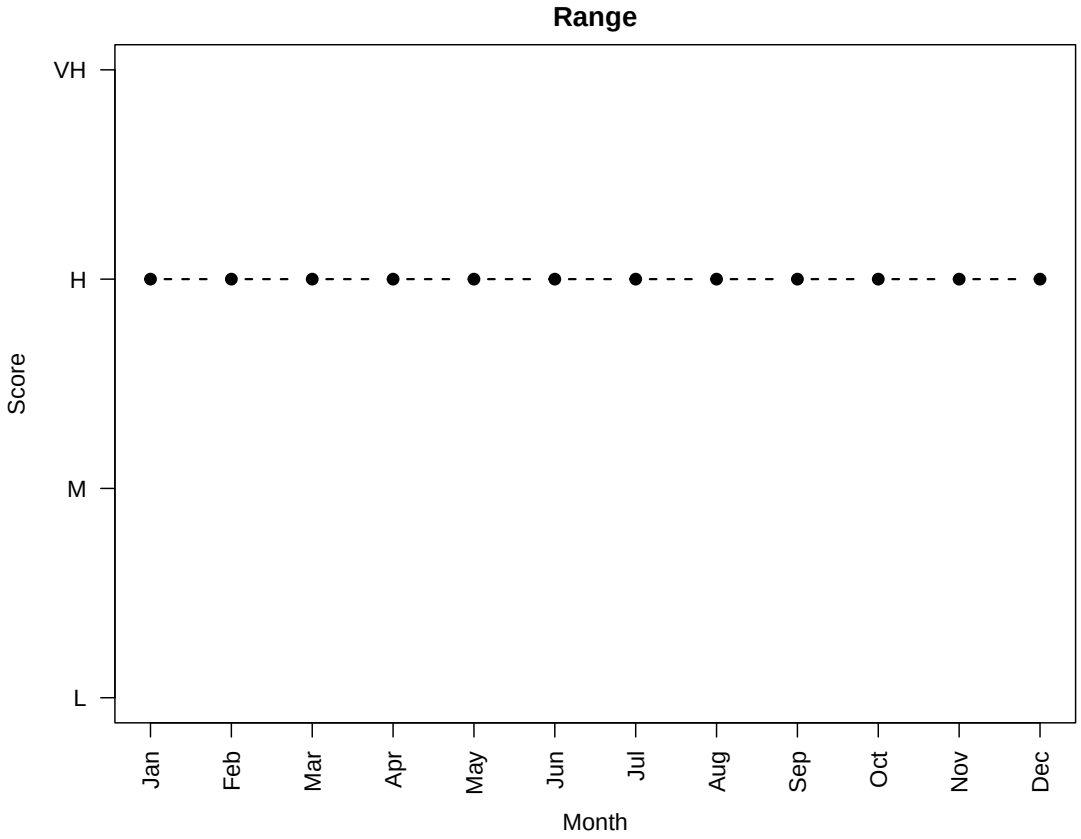


Figure 8: Timeseries of Total Exposure for Atlantic herring

Exposure due to Significant Factors

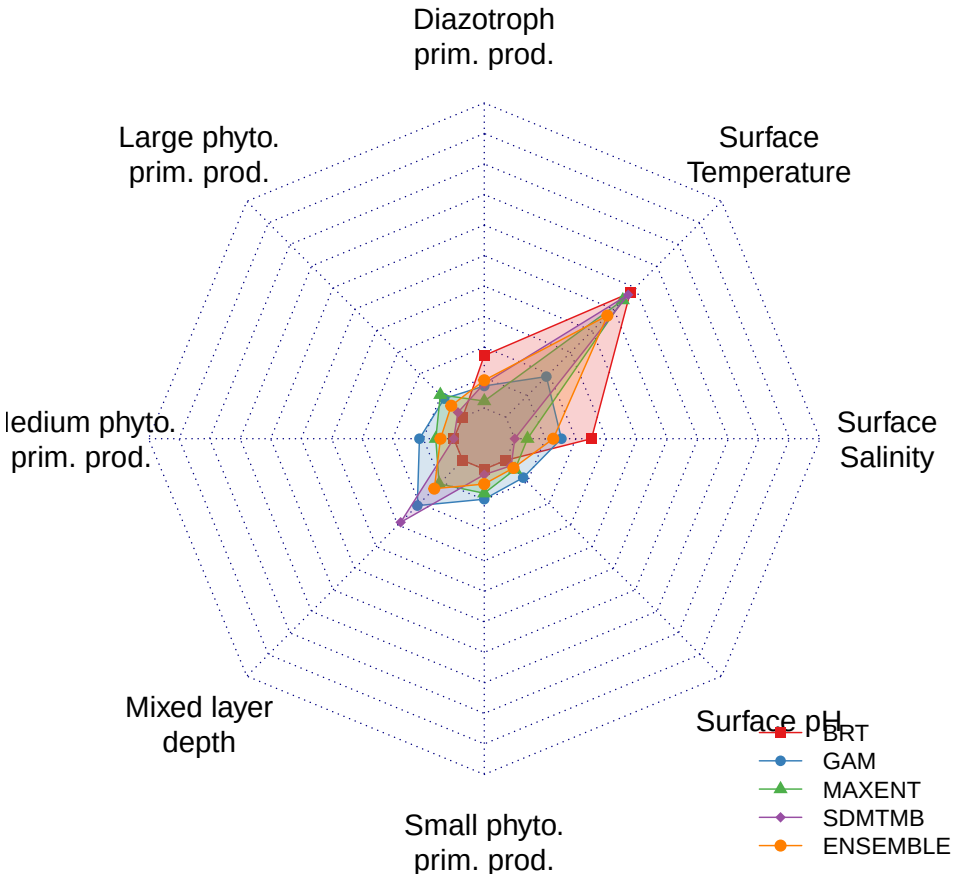


Figure 9: Dynamic Factors driving Atlantic herring distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Sea surface temperature, surface salinity, and mixed layer depth were all important drivers of herring distributions (Figure 9). Temperature and salinity have been linked to favorable habitats for herring (Suca et al. 2021) and increased mixing has been linked to decreased prey availability and increased mortality (Stevenson and Scott 2005). When these factors alone were used to calculate exposure, mean Exposure dropped to Moderate (1.72 +/- 0.83) (Figure 10). Exposure due to these variables was greatest in the Gulf of Maine and in the middle of the continental shelf (Figure 11).

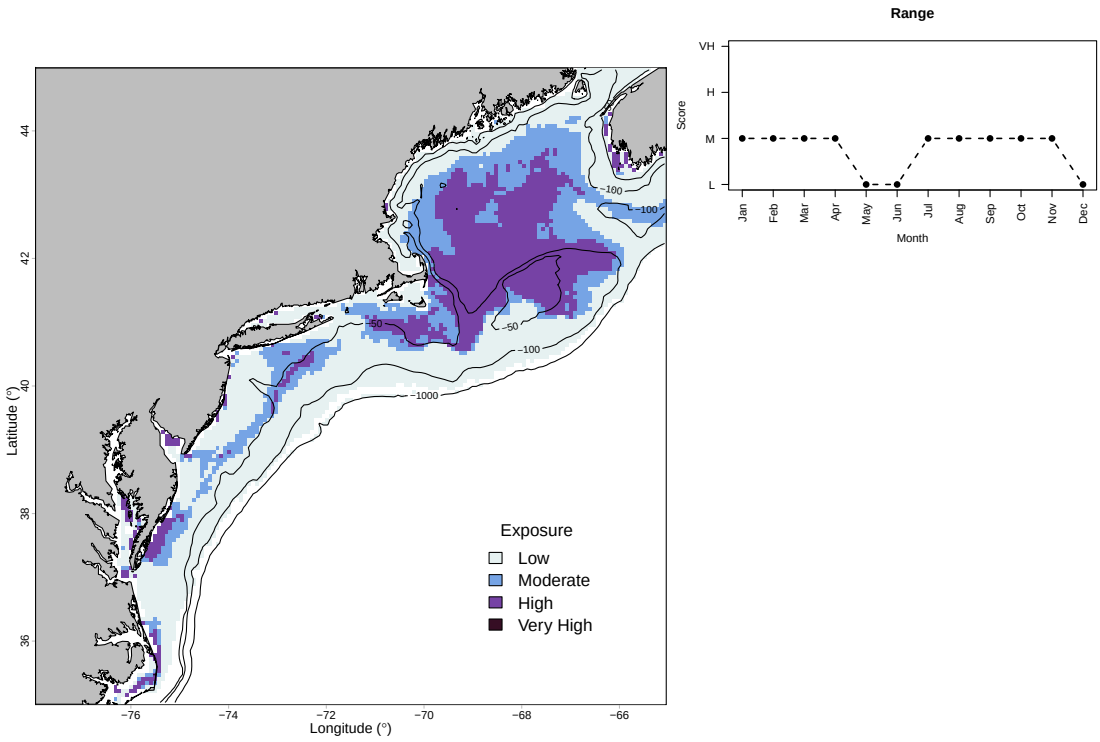


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Atlantic herring distributions

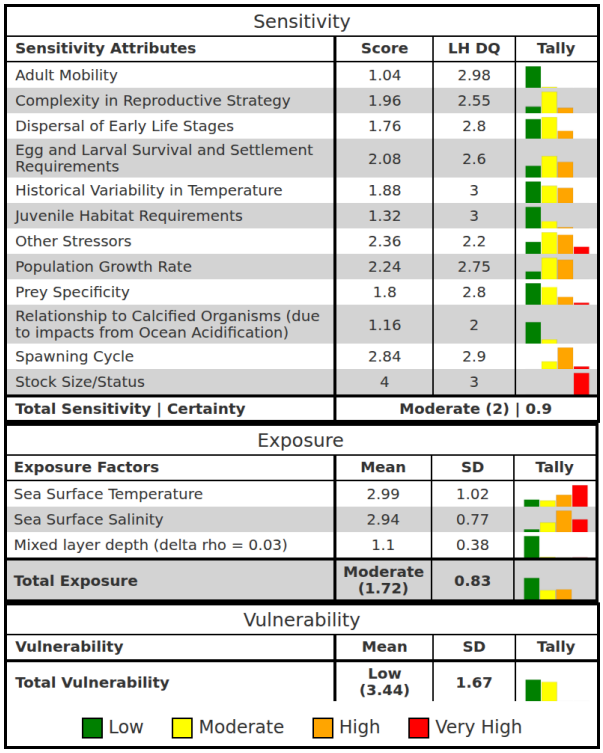


Figure 11: As in Figure 1 for Atlantic herring, but with only significant Exposure factors and the resulting Vulnerability Scores.

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